Nonparametric Models

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Empirical distribution

Definition 0.1 (Empirical distribution)**.** For observed values $x_1, \ldots, x_n$, the **empirical distribution** is the distribution of a value chosen at random from the sample, each of the n observations being equally likely: the probability that the value falls in a set A is

$$\frac{1}{n}\sum_{j=1}^{n} \mathbb{1}(x_j \in A).$$

Example 0.1 (Numerical example: empirical distribution)**.** Suppose the observed values are $4, 1, 7, 3$, so $n = 4$, and let $A = [2, 5)$. Two of the four values (4 and 3) lie in A, so the empirical distribution gives A probability $2/4 = 0.5$. Each observed value has probability $1/4$, and any set containing none of them has probability 0.

Remark (What "empirical" means). "Empirical" means computed from the observed data, not from a model or a population. An empirical quantity is the matching population quantity with the population distribution replaced by the empirical distribution. For example, the mean of the empirical distribution is the sample mean, and its CDF is the empirical CDF. Machine learning uses the same replacement to turn the risk of a prediction model into its *empirical risk*.
Because it needs only the data in hand, an empirical quantity can always be computed, and it serves as an estimate of the population quantity it mimics. The bootstrap rests on this replacement: it estimates the variability of a statistic by sampling from the empirical distribution (Hastie et al. 2009, sec. 7.11, p. 250; Efron and Tibshirani 1993).

Empirical CDF

Definition 0.2 (Empirical CDF)**.** For observed values $x_1, \ldots, x_n$, the **empirical cumulative distribution function** (**empirical CDF**) is the CDF of the empirical distribution, the function

$$\hat{\mathrm{F}}(t) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{j=1}^{n} I(x_j \leq t), \quad t \in \mathbb{R},$$

where $I(A)$ is the indicator function: $I(A) = 1$ if A holds, and $I(A) = 0$ otherwise.

Example 0.2 (Numerical example: empirical CDF)**.** Suppose the observed values are $4, 1, 7, 3$, so $n = 4$. Two of the four values (1 and 3) are at most 3.5, so:

$$\begin{aligned} \hat{F}(3.5) &= \frac{1}{4}(I(4 \leq 3.5) + I(1 \leq 3.5) + I(7 \leq 3.5) + I(3 \leq 3.5)) && \text{(definition of } \hat{F}) \\ &= \frac{1}{4}(0 + 1 + 0 + 1) && \text{(evaluate each indicator)} \\ &= 0.5 && \text{(arithmetic)} \end{aligned}$$

Similarly, $\hat{F}(0) = 0$, because no value is at most 0, and $\hat{F}(7) = 1$, because every value is at most 7.

Order statistics

Definition 0.3 (Order statistics)**.** For observed values $x_1, \ldots, x_n$, the **order statistics** are the same values sorted in increasing order, written

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}.$$

So $x_{(1)}$ is the smallest value and $x_{(n)}$ is the largest.

Example 0.3 (Numerical example: order statistics)**.** If the observed values are $4, 1, 7, 3$, then the order statistics are $x_{(1)} = 1$, $x_{(2)} = 3$, $x_{(3)} = 4$, and $x_{(4)} = 7$.

Sample quantiles

Definition 0.4 (Sample quantile function)**.** Given an empirical CDF $\hat{F}$, the **sample quantile function** (also called the **empirical quantile function**) is

$$\hat{Q}(p) \stackrel{\text{def}}{=} \inf\{t : \hat{F}(t) \geq p\}, \quad 0 < p \leq 1.$$

Definition 0.5 (Sample quantile)**.** For a given p, the value $\hat{Q}(p)$ of the sample quantile function is the **sample p quantile**, also called the **sample $100p$th percentile**.

The CDF F and the quantile function Q describe a probability distribution, while $\hat{F}$ and $\hat{Q}$ describe a sample, and serve as estimates of F and Q.

Theorem 0.1 (Order-statistics form of the sample quantile)**.** *If $x_{(1)} \leq \cdots \leq x_{(n)}$ are the order statistics and $i \in \{1, \ldots, n\}$, then $\hat{Q}(p) = x_{(i)}$ for every p with $(i-1)/n < p \leq i/n$. In particular, $\hat{Q}(i/n) = x_{(i)}$.*

Proof

Proof. Fix p with $(i-1)/n < p \leq i/n$.
For any $t < x_{(i)}$, the $n - i + 1$ values $x_{(i)}, \ldots, x_{(n)}$ all exceed t, so at most $i - 1$ values are at most t, and:

$$\begin{aligned} \hat{F}(t) &\leq \frac{i-1}{n} && \text{(at most } i-1 \text{ indicators equal 1)} \\ &< p && \text{(choice of } p\text{)} \end{aligned}$$

So no $t < x_{(i)}$ belongs to the set $\{t : \hat{F}(t) \geq p\}$.
At $t = x_{(i)}$, the i values $x_{(1)}, \ldots, x_{(i)}$ are all at most $x_{(i)}$, so:

$$\begin{aligned} \hat{F}\big(x_{(i)}\big) &\geq \frac{i}{n} && \text{(at least } i \text{ indicators equal 1)} \\ &\geq p && \text{(choice of } p\text{)} \end{aligned}$$

So $x_{(i)}$ belongs to the set, and it is the smallest member of the set, because no smaller t belongs to it. Therefore $\hat{Q}(p) = x_{(i)}$. □

Example 0.4 (Numerical example: sample quantiles)**.** Using the same data $4, 1, 7, 3$, with order statistics $1, 3, 4, 7$, Theorem 0.1 says the sample quantile function is a step function:

- $\hat{Q}(p) = 1$ for $0 < p \leq 1/4$;
- $\hat{Q}(p) = 3$ for $1/4 < p \leq 1/2$;
- $\hat{Q}(p) = 4$ for $1/2 < p \leq 3/4$;
- $\hat{Q}(p) = 7$ for $3/4 < p \leq 1$.

So $\hat{Q}(0.5) = 3$ and $\hat{Q}(0.9) = 7$. R's `quantile()` function computes this definition when called with `type = 1`:

```
quantile(c(4, 1, 7, 3), probs = c(0.5, 0.9), type = 1)
#> 50% 90%
#>   3   7
```

Remark 0.1 (Quantile conventions). Other sources, and other software defaults, define sample quantiles differently, mostly by interpolating between order statistics. R's `quantile()` offers nine definitions through its `type` argument, and its default (`type = 7`) interpolates, so `quantile(c(4, 1, 7, 3), 0.5)` returns 3.5, not 3. The usual sample median is another interpolated quantile: for an even number of observations, it averages the two middle order statistics. These notes use Definition 0.4, which always returns one of the observed values.

The empirical CDF and quantile function as generalized inverses

Both functions in Figure 1 are step functions, so neither is one-to-one. Each flat piece of one function corresponds to a jump of the other: for example, $\hat{F}(t) = 0.5$ for every $t \in [3, 4)$, and $\hat{Q}$ jumps from 3 to 4 at $p = 0.5$. So $\hat{Q}$ is a *generalized* inverse of $\hat{F}$, not an inverse in the usual sense: $\hat{Q}(\hat{F}(t)) \leq t$ for every $t \geq x_{(1)}$, with equality only at the observed values.

Efron, Bradley, and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Monographs on Statistics and Applied Probability 57. Chapman & Hall/CRC. https://doi.org/10.1201/9780429246593.

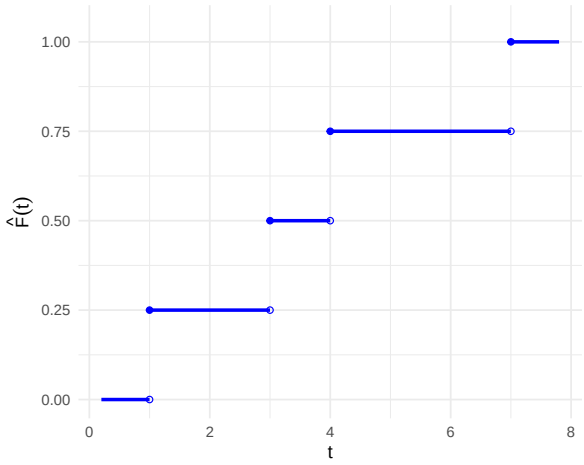
Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd ed. Springer. https://doi.org/10.1007/978-0-387-84858-7.

```r
x <- c(4, 1, 7, 3)
n <- length(x)
x_ord <- sort(x)
p_ord <- seq_len(n) / n
ecdf_x_padding <- 0.8
x_min <- min(x_ord) - ecdf_x_padding
x_max <- max(x_ord) + ecdf_x_padding

x_left <- c(x_min, x_ord)
x_right <- c(x_ord, x_max)
f_levels <- c(0, p_ord)

ecdf_pieces <- tibble::tibble(x = x_le
ecdf_open <- tibble::tibble(x = x_ord,
ecdf_closed <- tibble::tibble(x = x_or

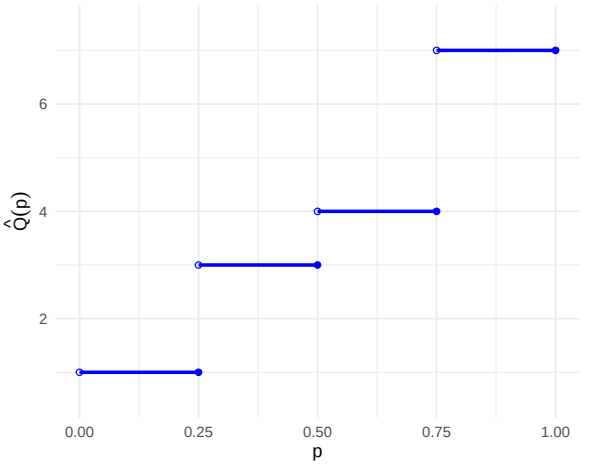
ggplot2::ggplot() +
  ggplot2::geom_segment(
    ggplot2::aes(x = x, xend = xend, y
    data = ecdf_pieces, linewidth = 1,
  ) +
  ggplot2::geom_point(
    ggplot2::aes(x = x, y = y), data =
  ) +
  ggplot2::geom_point(
    ggplot2::aes(x = x, y = y), data =
  ) +
  ggplot2::coord_cartesian(xlim = c(x_m
  ggplot2::labs(x = "t", y = expression
  ggplot2::theme_minimal()
```



(a) Empirical CDF, horizontal pieces only. Closed circles mark included endpoints; open circles mark excluded endpoints.

```r
eqf_y_padding <- 0.5
q_left <- c(0, p_ord[-n])
q_right <- p_ord
eqf_pieces <- tibble::tibble(x = q_left, xend = q_righ

ggplot2::ggplot(eqf_pieces) +
  ggplot2::geom_segment(
    ggplot2::aes(x = x, xend = xend, y = y, yend = y),
    linewidth = 1, color = "blue"
  ) +
  ggplot2::geom_point(ggplot2::aes(x = x, y = y), shap
  ggplot2::geom_point(
    ggplot2::aes(x = xend, y = y), shape = 19, color =
  ) +
  ggplot2::coord_cartesian(
    xlim = c(0, 1),
    ylim = c(min(x_ord) - eqf_y_padding, max(x_ord) +
  ) +
  ggplot2::labs(x = "p", y = expression(hat(Q)(p))) +
  ggplot2::theme_minimal()
```



(b) Sample quantile function, horizontal pieces only. Open circles mark excluded left endpoints; closed circles mark included right endpoints.

Figure 1: The empirical CDF and the sample quantile function for the data 4, 1, 7, 3 (order statistics 1, 3, 4, 7).